- Grover's algorithm
- Shor's algorithm
- Qiskit programming
- Laboratory experiments

The emphasis is on understanding the underlying physics while developing practical programming skills using IBM Qiskit.

Worked Example

Suppose a classical bit stores the value

1.

Its state is fixed until changed.

A qubit, however, may be prepared as

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Before measurement, the qubit simultaneously possesses amplitudes corresponding to both basis states.

Upon measurement,

$$P(0) = \frac{1}{2}, \qquad P(1) = \frac{1}{2}.$$

Thus, the measurement yields either outcome with equal probability.

Key Takeaways

- Quantum computing exploits quantum mechanical principles for computation.
- Qubits differ fundamentally from classical bits.
- Superposition and entanglement enable quantum computational advantages.
- Quantum computing has applications in science, engineering, optimization and cryptography.
- Modern quantum computers are already accessible through cloud platforms such as IBM Quantum.

Summary

Quantum computing represents one of the most promising technological advances of the twenty-first century. By utilizing quantum mechanical phenomena such as superposition and entanglement, quantum computers can solve selected computational problems much more efficiently than classical computers. This course provides the mathematical foundations, physical principles, quantum algorithms, and practical programming skills required to understand and develop quantum technologies.

Review Questions

1. What is a qubit?
2. List four differences between classical and quantum computation.
3. Explain the concept of superposition.
4. Name four physical implementations of qubits.
5. Mention three important applications of quantum computing.

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
3. David McMahon, *Quantum Computing Explained*.
4. IBM Quantum Documentation.
5. Qiskit Documentation.

Chapter 2

Linear Algebra Review for Quantum Mechanics

Learning Objectives

After completing this lecture, students will be able to:

- Understand why linear algebra forms the mathematical foundation of quantum mechanics.
- Represent vectors and matrices used in quantum computing.
- Perform basic vector and matrix operations.
- Understand linear independence and basis vectors.
- Apply these concepts to quantum state representation.

Prerequisites

Knowledge of elementary algebra, matrices, and complex numbers.

Introduction

Linear algebra is the language of quantum mechanics. Every quantum state is represented by a vector, while every observable and quantum gate is represented by a matrix (operator). Unlike classical mechanics, where physical quantities are described using numbers and functions, quantum mechanics describes systems in abstract vector spaces called Hilbert spaces.

Consequently, understanding vectors, matrices, and their operations is essential before studying quantum states, quantum gates, and quantum algorithms.

Scalars and Vectors

A **scalar** is a quantity described by a single number.

Examples include

$$5, \qquad -2, \qquad \pi, \qquad 3+2i.$$

A **vector** possesses both magnitude and direction and is represented as an ordered collection of numbers.

For example,

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

In quantum mechanics, vectors generally contain complex numbers.

Vector Operations

Two vectors can be added if they have the same dimension.

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

Scalar multiplication changes the magnitude of a vector.

$$3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}.$$

These operations satisfy closure, associativity, and distributive laws, making vectors elements of a vector space.

Matrices

A matrix is a rectangular arrangement of numbers.

A general matrix is written as

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Matrices are used extensively in quantum mechanics because they represent physical operations acting on quantum states.

Common matrix types include

- Square matrices

- Identity matrices
- Diagonal matrices
- Hermitian matrices
- Unitary matrices

The identity matrix is

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

which leaves every vector unchanged.

Matrix Operations

Two matrices of equal dimensions may be added element-wise.

For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}.$$

Matrix multiplication is defined only when the number of columns of the first matrix equals the number of rows of the second.

For matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix},$$

their product is

$$AB = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}.$$

Unlike ordinary numbers,

$$AB \neq BA$$

in general. This property is central to quantum mechanics and leads to the concept of non-commuting observables.

Vector Spaces

A vector space is a collection of vectors that satisfies the following properties:

- Closure under addition
- Closure under scalar multiplication
- Existence of a zero vector
- Existence of additive inverses
- Associative and distributive laws

Quantum states exist in complex vector spaces called Hilbert spaces, which additionally possess an inner product.

Linear Independence and Basis

A set of vectors is said to be linearly independent if none of the vectors can be written as a linear combination of the others.

For example,

$$\begin{bmatrix}1\\0\end{bmatrix}, \qquad \begin{bmatrix}0\\1\end{bmatrix}$$

form an independent set.

These vectors constitute the standard basis for a two-dimensional vector space.

In quantum computing, these basis vectors correspond to the computational basis states

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \qquad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}.$$

Every single-qubit quantum state is expressed as a linear combination of these basis vectors.

Worked Example

Consider the vectors

$$\mathbf{u} = \begin{bmatrix}2\\1\end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix}3\\4\end{bmatrix}.$$

Their sum is

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix}5\\5\end{bmatrix}.$$

Now consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Multiplying the matrix by the vector,

$$A\mathbf{u} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

This illustrates how matrices transform vectors. Quantum gates similarly transform quantum states.

Connection to Quantum Computing

Linear algebra provides the mathematical tools for representing:

- Quantum states as vectors.
- Quantum gates as matrices.
- Measurements as operators.
- Quantum evolution as matrix transformations.
- Multi-qubit systems using tensor products.

Every quantum algorithm ultimately consists of successive matrix operations acting on state vectors.

Key Takeaways

- Quantum mechanics is fundamentally built upon linear algebra.
- Quantum states are vectors in complex vector spaces.
- Quantum gates are represented by matrices.
- Matrix multiplication is generally non-commutative.
- Computational basis states arise naturally from the standard basis vectors.

Summary

Linear algebra provides the mathematical framework for quantum computing. Vectors represent quantum states, while matrices describe physical transformations and quantum gates. Concepts such as vector spaces, basis vectors, linear independence, and matrix multiplication form the foundation for understanding Hilbert spaces, quantum operators, and quantum algorithms. Mastery of these tools is essential for all subsequent topics in this course.

Review Questions

1. Why is linear algebra important in quantum mechanics?
2. Define a vector space.
3. Explain the difference between scalar multiplication and matrix multiplication.
4. Why is matrix multiplication generally non-commutative?
5. Write the computational basis vectors for a single qubit.
6. What is the significance of the identity matrix?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. J. J. Sakurai, *Modern Quantum Mechanics*.
3. David A. McMahon, *Quantum Computing Explained*.
4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
5. IBM Quantum Documentation.

Chapter 3

Vectors, Inner Products and Vector Spaces

Learning Objectives

After studying this lecture, students will be able to

- Represent vectors in complex vector spaces.
- Define inner products and vector norms.
- Understand orthogonality and orthonormal basis vectors.
- Explain the concept of vector spaces and Hilbert spaces.
- Apply these ideas to quantum state representation.

Prerequisites

Basic knowledge of matrices, vectors, and complex numbers.

Introduction

Quantum mechanics is formulated using the mathematics of vector spaces. Every quantum state is represented by a vector in a complex vector space called a *Hilbert space*. Physical observables are represented by linear operators acting on these vectors. Consequently, concepts such as inner products, orthogonality, and basis vectors are fundamental to quantum computing.

Vectors in Quantum Mechanics

A vector is an ordered collection of numbers that may be real or complex. In quantum mechanics, vectors are written as column matrices. For example,

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}.$$

A single-qubit quantum state is represented as

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

where α and β are complex probability amplitudes satisfying

$$|\alpha|^2 + |\beta|^2 = 1.$$

This normalization ensures that the total probability of all possible measurement outcomes equals one.

Inner Product

The inner product measures the overlap between two vectors. For vectors

$$|u\rangle = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \qquad |v\rangle = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

their inner product is

$$\langle u|v\rangle = u_1^* v_1 + u_2^* v_2,$$

where * denotes complex conjugation.
The inner product satisfies the following properties:

- Linearity
- Conjugate symmetry
- Positive definiteness

If

$$\langle u|v\rangle = 0,$$

the vectors are said to be **orthogonal**.

Norm of a Vector

The magnitude or length of a vector is called its norm and is defined by

$$||v|| = \sqrt{\langle v|v\rangle}.$$

For the vector

$$|v\rangle = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

the norm is

$$||v|| = \sqrt{3^2 + 4^2} = 5.$$

Normalized vectors satisfy

$$||v|| = 1.$$

All quantum state vectors must be normalized.

Basis Vectors

A basis is a collection of linearly independent vectors that spans the entire vector space.

For a single qubit, the computational basis consists of

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

These basis vectors satisfy

$$\langle 0|0\rangle = 1, \qquad \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = 0.$$

Thus, they form an orthonormal basis.

Any qubit can be expressed as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Vector Spaces

A vector space is a collection of vectors satisfying the following conditions:

- Closure under vector addition.

- Closure under scalar multiplication.
- Existence of a zero vector.
- Existence of additive inverses.
- Associative and distributive properties.

Examples include

- Two-dimensional Euclidean space.
- Three-dimensional Cartesian space.
- Complex vector spaces used in quantum mechanics.

Hilbert Space

A Hilbert space is a complete vector space equipped with an inner product.

Quantum mechanics requires Hilbert spaces because they allow

- normalization of quantum states,
- computation of probabilities,
- orthogonal basis states,
- expansion of arbitrary quantum states,
- representation of quantum operators.

For a single qubit, the Hilbert space is two-dimensional. Two qubits require a four-dimensional Hilbert space, while an n-qubit system requires a Hilbert space of dimension

$$2^n.$$

This exponential growth is one of the major sources of quantum computational power.

Worked Example

Consider the vectors

$$|u\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |v\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The inner product is

$$\langle u|v\rangle = (1)(0) + (0)(1) = 0.$$

Therefore,

$$|u\rangle \perp |v\rangle.$$

Now consider the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Its norm is

$$||\psi||^2 = \frac{1}{2} + \frac{1}{2} = 1,$$

showing that it is a valid normalized quantum state.

Applications in Quantum Computing

Vectors and inner products are used to

- represent qubits,
- calculate measurement probabilities,
- determine orthogonal quantum states,
- construct quantum algorithms,
- describe quantum gates as linear transformations.

Nearly every quantum algorithm relies on operations performed within Hilbert spaces.

Key Takeaways

- Quantum states are vectors in complex Hilbert spaces.
- Inner products determine probabilities and orthogonality.
- Normalized vectors represent valid quantum states.
- Computational basis vectors form an orthonormal basis.
- Hilbert spaces provide the mathematical framework for quantum mechanics.

Summary

Vectors, inner products, and vector spaces form the mathematical backbone of quantum mechanics. Quantum states are represented as normalized vectors in Hilbert spaces, while inner products quantify their similarity and determine measurement probabilities. The computational basis vectors $|0\rangle$ and $|1\rangle$ provide the foundation for representing qubits, and these concepts will be used extensively when studying quantum gates, operators, and algorithms.

Review Questions

1. What is an inner product?
2. Define the norm of a vector.
3. What is meant by orthogonal vectors?
4. Why must quantum state vectors be normalized?
5. What is a Hilbert space?
6. Why does an n-qubit system require a 2^n-dimensional vector space?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. J. J. Sakurai, *Modern Quantum Mechanics*.
3. David McMahon, *Quantum Computing Explained*.
4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
5. IBM Quantum Documentation.

Chapter 4

Matrix Algebra and Linear Transformations

Learning Objectives

After completing this lecture, students will be able to

- Understand matrix algebra and its properties.
- Perform matrix addition and multiplication.
- Explain linear transformations.
- Identify special matrices used in quantum mechanics.
- Relate matrix operations to quantum gates.

Prerequisites

Knowledge of vectors, vector spaces, and inner products.

Introduction

Matrices are fundamental mathematical objects in quantum mechanics. Every quantum gate, observable, and state transformation is represented by a matrix acting on a vector. Unlike ordinary numbers, matrices generally do not commute under multiplication, leading to many uniquely quantum phenomena. Understanding matrix algebra is therefore essential for studying quantum operators and quantum circuits.

Matrices

A matrix is a rectangular array of numbers arranged in rows and columns.

A general $m \times n$ matrix is written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Quantum computing primarily uses square matrices because they represent operations on quantum states.

Basic Matrix Operations

Matrix Addition

Matrices of identical dimensions are added element-wise.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}.$$

Scalar Multiplication

Multiplying every element by a constant gives

$$2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

Matrix Multiplication

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix},$$

then

$$AB = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}.$$

Matrix multiplication follows the row-by-column rule and is generally non-commutative:

$$AB \neq BA.$$

This property plays an important role in quantum mechanics.

Special Matrices

Frequently used matrices include

- Identity matrix
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
- Zero matrix
$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$
- Diagonal matrix
$$D = \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix}$$
- Transpose
$$A^T$$
obtained by interchanging rows and columns.
- Complex conjugate
$$A^*$$
obtained by replacing every complex number with its conjugate.
- Conjugate transpose (Hermitian conjugate)
$$A^\dagger = (A^*)^T.$$

The Hermitian conjugate is extensively used in quantum mechanics.

Linear Transformations

A linear transformation maps one vector into another while preserving vector addition and scalar multiplication.

If T is a linear transformation,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v}),$$

where a and b are scalars.

Every linear transformation can be represented by a matrix.

For a matrix A acting on a vector $\mathbf{x}$,

$$\mathbf{y} = A\mathbf{x}.$$

Thus, matrices describe how vectors are transformed from one state to another.

Linear Transformations in Quantum Computing

Quantum gates are examples of linear transformations.

For example, the Pauli-X gate is

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Applying it to

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

gives

$$X|0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle.$$

Similarly,

$$X|1\rangle = |0\rangle.$$

Thus, the Pauli-X gate performs a quantum bit-flip operation.

Worked Example

Consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

and the vector

$$\mathbf{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Then,

$$A\mathbf{x} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}.$$

The matrix has transformed the original vector into a new vector, illustrating the action of a linear operator.

Applications in Quantum Computing

Matrix algebra is used to

- represent quantum gates,
- describe state evolution,
- perform quantum measurements,
- construct quantum circuits,
- simulate quantum algorithms.

Nearly every computation performed on a quantum computer consists of a sequence of matrix transformations acting on quantum states.

Key Takeaways

- Matrices represent linear operators in quantum mechanics.
- Matrix multiplication is generally non-commutative.
- Linear transformations preserve vector addition and scalar multiplication.
- Quantum gates are unitary matrix transformations.
- Matrix algebra forms the computational basis of quantum computing.

Summary

Matrix algebra provides the mathematical framework for describing quantum operations. Every quantum gate is a linear transformation represented by a matrix acting on a state vector. Important concepts such as matrix multiplication, transpose, conjugate transpose, and identity matrices are essential for understanding quantum operators, state evolution, and quantum circuits. These ideas prepare the foundation for studying Hermitian operators and quantum measurements.

Review Questions

1. Define matrix multiplication.
2. Why is matrix multiplication generally non-commutative?
3. What is a linear transformation?
4. Define the Hermitian conjugate of a matrix.
5. Explain why quantum gates are represented by matrices.
6. Demonstrate how the Pauli-X gate transforms $|0\rangle$ into $|1\rangle$.

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. J. J. Sakurai, *Modern Quantum Mechanics*.
3. David McMahon, *Quantum Computing Explained*.
4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
5. IBM Quantum Documentation.

Chapter 5

Heisenberg Matrix Representation

Learning Objectives

After completing this lecture, students will be able to

- Understand Heisenberg's matrix formulation of quantum mechanics.
- Explain the concept of observables as matrices.
- Distinguish between Schrödinger and Heisenberg pictures.
- Interpret matrix elements as measurable physical quantities.
- Appreciate the importance of matrix mechanics in quantum computing.

Prerequisites

Knowledge of vectors, matrices, linear transformations, and basic quantum concepts.

Introduction

In 1925, Werner Heisenberg introduced the first successful mathematical formulation of quantum mechanics, known as *matrix mechanics*. Instead of describing particles by trajectories, Heisenberg represented measurable physical quantities such as position, momentum, and angular momentum using matrices. This revolutionary idea formed the foundation of modern quantum mechanics and later became indispensable in quantum computing, where quantum states and gates are naturally represented using matrices.

Matrix Representation of Observables

In classical mechanics, physical quantities are represented by numerical variables. In quantum mechanics, measurable quantities (called *observables*) are represented by operators, which are expressed as matrices.

Examples include

$$\hat{x}, \qquad \hat{p}, \qquad \hat{H},$$

representing position, momentum, and energy (Hamiltonian), respectively.

For a two-level quantum system, an observable may be represented by

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

The eigenvalues of this matrix correspond to the possible outcomes of a measurement.

State Vectors and Operators

A quantum state is represented by a column vector,

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

When an operator acts on the state,

$$A|\psi\rangle,$$

a new quantum state is obtained. Thus, operators describe physical transformations and measurements.

Heisenberg Picture

Quantum mechanics can be formulated in two equivalent ways.

- In the **Schrödinger picture**, state vectors evolve with time while operators remain constant.
- In the **Heisenberg picture**, operators evolve with time while state vectors remain fixed.

The time evolution of an operator in the Heisenberg picture is given by

$$A_H(t) = U^\dagger(t)\, A\, U(t),$$

where

$$U(t) = e^{-iHt/\hbar}$$

is the unitary time-evolution operator and H is the Hamiltonian.

Although the mathematical descriptions differ, both formulations predict identical physical results.

Commutation Relations

One of Heisenberg's major contributions was recognizing that many physical quantities do not commute.

For two operators A and B, the commutator is defined as

$$[A, B] = AB - BA.$$

If

$$[A, B] = 0,$$

the observables are simultaneously measurable.

A famous example is the position and momentum operators,

$$[\hat{x}, \hat{p}] = i\hbar,$$

which leads directly to Heisenberg's uncertainty principle.

Pauli Matrices

Matrix mechanics is illustrated by the Pauli matrices, which describe the spin of a spin-$\frac{1}{2}$ particle.

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

These matrices are Hermitian and play a central role in quantum computing, where they correspond to the quantum X, Y, and Z gates.

Worked Example

Consider the Pauli-X operator,

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

acting on the computational basis state

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The transformed state is

$$X|0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle.$$

Similarly,

$$X|1\rangle = |0\rangle.$$

Thus, the matrix operator performs a bit-flip operation, demonstrating how quantum gates are naturally represented using Heisenberg's matrix formalism.

Applications in Quantum Computing

Matrix representation is used to

- describe quantum gates,
- model quantum measurements,
- calculate expectation values,
- simulate quantum circuits,
- implement quantum algorithms using linear algebra.

Modern quantum software frameworks such as Qiskit internally represent quantum operations using matrices and state vectors.

Key Takeaways

- Heisenberg introduced matrix mechanics in 1925.
- Quantum observables are represented by matrices (operators).
- Matrix eigenvalues correspond to measurable quantities.
- The Heisenberg and Schrödinger pictures are mathematically equivalent.
- Matrix mechanics provides the mathematical foundation for quantum computing.

Summary

Heisenberg's matrix representation transformed quantum mechanics by replacing classical variables with matrix operators. Observables such as position, momentum, and spin are represented by Hermitian matrices that act on quantum state vectors. The non-commutative nature of these matrices explains fundamental quantum phenomena, including the uncertainty principle. Today, matrix mechanics remains the language of quantum computing, where quantum gates, measurements, and algorithms are expressed entirely through matrix operations.

Review Questions

1. What is Heisenberg's matrix representation?
2. How are observables represented in quantum mechanics?
3. Distinguish between the Schrödinger and Heisenberg pictures.
4. Define the commutator of two operators.
5. State the commutation relation between position and momentum.
6. Why are the Pauli matrices important in quantum computing?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information.*
2. J. J. Sakurai, *Modern Quantum Mechanics.*
3. David Griffiths, *Introduction to Quantum Mechanics.*

4. David McMahon, *Quantum Computing Explained*.

5. IBM Quantum Documentation.

Chapter 6

Quantum States and Hilbert Space I

Learning Objectives

After completing this lecture, students will be able to

- Define a quantum state.
- Understand the postulate of state representation.
- Explain the concept of Hilbert space.
- Represent qubit states using Dirac notation.
- Interpret normalization and probability amplitudes.

Prerequisites

Knowledge of vectors, inner products, vector spaces, and matrix algebra.

Introduction

In classical mechanics, the state of a particle is completely specified by its position and momentum. In contrast, quantum mechanics represents the state of a physical system by a vector in an abstract mathematical space known as a *Hilbert space*. This vector contains all the information required to predict the outcomes of measurements.

Quantum states form the foundation of quantum computing because every qubit, quantum register, and quantum algorithm is ultimately described using state vectors in Hilbert space.

Quantum State

A **quantum state** completely specifies the physical condition of a quantum system.

According to the first postulate of quantum mechanics,

> *The state of an isolated quantum system is represented by a normalized vector in a complex Hilbert space.*

This state is written using Dirac's bra-ket notation as

$$|\psi\rangle.$$

For a single qubit,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

- α and β are complex probability amplitudes,
- $|0\rangle$ and $|1\rangle$ are basis states.

The normalization condition is

$$|\alpha|^2 + |\beta|^2 = 1.$$

Computational Basis States

The simplest Hilbert space is two-dimensional and represents a single qubit.

Its basis vectors are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

These vectors satisfy

$$\langle 0|0\rangle = 1, \qquad \langle 1|1\rangle = 1,$$

and

$$\langle 0|1\rangle = 0.$$

Thus, they form an orthonormal basis.

Any single-qubit state can be expressed as a linear combination of these two basis vectors.

Hilbert Space

A Hilbert space is a complete complex vector space equipped with an inner product.

It possesses the following properties:

- Vector addition
- Scalar multiplication
- Inner product
- Norm
- Completeness

Unlike ordinary Euclidean space, Hilbert spaces may have infinitely many dimensions. In quantum computing, however, finite-dimensional Hilbert spaces are commonly used.

For an n-qubit system, the Hilbert space has dimension

$$2^n.$$

Thus,

1 qubit	2 dimensions
2 qubits	4 dimensions
3 qubits	8 dimensions
n qubits	2^n dimensions

This exponential growth enables quantum computers to process large state spaces efficiently.

Normalization

A valid quantum state must have unit norm.

If

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

then

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

The quantities

$$|\alpha|^2 \quad \text{and} \quad |\beta|^2$$

represent the probabilities of obtaining the outcomes $|0\rangle$ and $|1\rangle$, respectively, upon measurement.

Superposition of States

One of the most remarkable features of quantum mechanics is superposition.

Instead of existing exclusively in one basis state, a qubit may exist in a combination of basis states.

For example,

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

When measured,

$$P(0) = \frac{1}{2}, \qquad P(1) = \frac{1}{2}.$$

The system does not possess a definite classical value before measurement; instead, it is described by probability amplitudes.

Worked Example

Consider the state

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle.$$

Check normalization:

$$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9}{25} + \frac{16}{25} = 1.$$

Hence, this is a valid quantum state.

The probabilities of measurement are

$$P(0) = \frac{9}{25} = 0.36,$$

$$P(1) = \frac{16}{25} = 0.64.$$

Thus, the qubit is more likely to be measured in the state $|1\rangle$.

Applications in Quantum Computing

Quantum state representation is essential for

- describing qubits,
- constructing quantum registers,
- analyzing quantum gates,
- performing quantum measurements,
- implementing quantum algorithms.

Every quantum computation begins with preparing an initial quantum state in Hilbert space.

Key Takeaways

- A quantum state is represented by a normalized vector.
- Hilbert space is the mathematical space of quantum states.
- The computational basis consists of $|0\rangle$ and $|1\rangle$.
- Probability amplitudes determine measurement outcomes.
- Hilbert-space dimension doubles with every additional qubit.

Summary

Quantum states provide the complete description of isolated quantum systems. These states are represented by normalized vectors in Hilbert space using Dirac notation. The computational basis vectors $|0\rangle$ and $|1\rangle$ span the Hilbert space of a single qubit, while larger quantum systems require exponentially larger Hilbert spaces. Understanding quantum states and Hilbert spaces is essential for studying superposition, entanglement, quantum gates, and quantum algorithms.

Review Questions

1. Define a quantum state.
2. State the first postulate of quantum mechanics.

3. What is a Hilbert space?
4. Why must a quantum state be normalized?
5. What are the computational basis states?
6. Why does an n-qubit system require a 2^n-dimensional Hilbert space?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. David J. Griffiths, *Introduction to Quantum Mechanics*.
3. J. J. Sakurai, *Modern Quantum Mechanics*.
4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
5. IBM Quantum Documentation.

Chapter 7

Quantum States and Hilbert Space II

Learning Objectives

After completing this lecture, students will be able to

- Represent multi-qubit quantum states.
- Understand tensor (Kronecker) products.
- Explain orthonormal bases in Hilbert space.
- Distinguish pure and mixed quantum states.
- Appreciate the exponential growth of Hilbert space.

Prerequisites

Knowledge of vectors, inner products, Dirac notation, and single-qubit Hilbert spaces.

Introduction

In the previous lecture, we learned that a single qubit is represented by a normalized vector in a two-dimensional Hilbert space. Practical quantum computers, however, operate on many qubits simultaneously. The mathematical description of such systems requires larger Hilbert spaces formed using the *tensor product*. This lecture introduces multi-qubit state representation and explains why Hilbert spaces grow exponentially with the number of qubits.

Tensor Product of States

The state of a composite quantum system is obtained using the tensor (Kronecker) product of individual qubit states.

If

$$|a\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \qquad |b\rangle = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix},$$

then

$$|a\rangle \otimes |b\rangle = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{bmatrix}.$$

The tensor product combines individual Hilbert spaces into a larger Hilbert space capable of describing composite quantum systems.

Two-Qubit Basis States

The computational basis for two qubits consists of four orthonormal states:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

Their vector representations are

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

$$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

These four vectors form an orthonormal basis for a four-dimensional Hilbert space.

General Multi-Qubit State

A general two-qubit state is

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1.$$

Similarly, an n-qubit quantum state contains 2^n probability amplitudes.

Dimension of Hilbert Space

Every additional qubit doubles the size of the Hilbert space.

Number of Qubits	Hilbert Space Dimension
1	2
2	4
3	8
4	16
5	32
n	2^n

This exponential scaling allows quantum computers to represent enormous state spaces using relatively few physical qubits.

Orthonormal Basis

The computational basis vectors satisfy

$$\langle i|j\rangle = \delta_{ij},$$

where δ_{ij} is the Kronecker delta,

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

An orthonormal basis simplifies calculations of probabilities, measurements, and quantum gate operations.

Pure and Mixed States

A **pure state** is described by a single normalized state vector,

$$|\psi\rangle.$$

For example,

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

is a pure state.

A **mixed state** represents a statistical mixture of different quantum states and cannot be described by a single state vector alone. Mixed states are represented using the density matrix, which will be studied in a later lecture.

Pure states exhibit complete quantum coherence, whereas mixed states arise from uncertainty or interaction with the environment.

Worked Example

Construct the two-qubit state

$$|0\rangle \otimes |1\rangle.$$

Since

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

their tensor product is

$$|01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Thus, the composite state belongs to a four-dimensional Hilbert space.

Applications in Quantum Computing

Multi-qubit Hilbert spaces are essential for

- quantum registers,
- quantum algorithms,
- entanglement,

- quantum error correction,
- quantum communication.

The computational advantage of quantum computers arises from manipulating vectors in these exponentially growing Hilbert spaces.

Key Takeaways

- Composite quantum systems are described using tensor products.
- Two qubits require a four-dimensional Hilbert space.
- An n-qubit system requires 2^n basis states.
- Computational basis vectors are orthonormal.
- Pure states differ fundamentally from mixed states.

Summary

Hilbert spaces provide the mathematical framework for describing multi-qubit quantum systems. By combining single-qubit Hilbert spaces using the tensor product, larger state spaces are constructed whose dimensions increase exponentially with the number of qubits. These spaces enable the representation of complex quantum states, including superposition and, later, entanglement. Understanding multi-qubit Hilbert spaces is essential for analyzing quantum circuits and algorithms.

Review Questions

1. What is the tensor product?
2. List the computational basis states of a two-qubit system.
3. Why does the Hilbert-space dimension double with each added qubit?
4. Define an orthonormal basis.
5. Distinguish between pure and mixed quantum states.
6. How many basis states are required for a five-qubit system?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.

2. David J. Griffiths, *Introduction to Quantum Mechanics*.

3. J. J. Sakurai, *Modern Quantum Mechanics*.

4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.

5. IBM Quantum Documentation.

Chapter 8

Bloch Sphere Representation

Learning Objectives

After completing this lecture, students will be able to

- Understand the Bloch sphere as a geometrical representation of a qubit.
- Represent arbitrary single-qubit states using spherical coordinates.
- Identify important quantum states on the Bloch sphere.
- Explain quantum gate operations as rotations.
- Appreciate the limitations of the Bloch sphere.

Prerequisites

Knowledge of qubits, Hilbert spaces, complex probability amplitudes, and Dirac notation.

Introduction

Although a qubit is mathematically represented by a two-dimensional complex vector, it is often difficult to visualize its state. The *Bloch sphere* provides an elegant geometrical representation of every pure single-qubit state as a point on the surface of a unit sphere. This visualization is widely used in quantum computing because quantum gates can be interpreted as rotations of the state vector on the sphere.

General Qubit State

A single qubit is represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Since an overall phase factor has no physical significance, every pure qubit state can be expressed in the form

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

where

- $0 \leq \theta \leq \pi$ is the polar angle,
- $0 \leq \phi < 2\pi$ is the azimuthal angle.

Thus, every pure qubit corresponds to a unique point on the Bloch sphere.

Bloch Sphere Coordinates

The Cartesian coordinates of the Bloch vector are

$$x = \sin\theta\cos\phi,$$

$$y = \sin\theta\sin\phi,$$

$$z = \cos\theta.$$

Since

$$x^2 + y^2 + z^2 = 1,$$

the state lies on the surface of a unit sphere.

Important Quantum States

Several commonly used quantum states occupy special positions on the Bloch sphere.

State	Location
$\|0\rangle$	North Pole ($+z$)
$\|1\rangle$	South Pole ($-z$)
$\frac{\|0\rangle+\|1\rangle}{\sqrt{2}}$	$+x$ axis
$\frac{\|0\rangle-\|1\rangle}{\sqrt{2}}$	$-x$ axis
$\frac{\|0\rangle+i\|1\rangle}{\sqrt{2}}$	$+y$ axis
$\frac{\|0\rangle-i\|1\rangle}{\sqrt{2}}$	$-y$ axis

These states frequently appear in quantum algorithms and quantum gate operations.

Quantum Gates as Rotations

Single-qubit quantum gates rotate the Bloch vector about one of the coordinate axes.

Examples include

- Pauli-X gate: rotation about the x-axis by π.
- Pauli-Y gate: rotation about the y-axis by π.
- Pauli-Z gate: rotation about the z-axis by π.
- Hadamard gate: rotates basis states between the z and x axes.

Therefore, quantum computation may be viewed as a sequence of controlled rotations on the Bloch sphere.

Worked Example

Consider the state

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Comparing with the general expression,

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

gives

$$\theta = \frac{\pi}{2}, \qquad \phi = 0.$$

Hence,

$$x = 1, \qquad y = 0, \qquad z = 0.$$

Thus, the state lies on the positive x-axis of the Bloch sphere.
Similarly,

$$|0\rangle$$

corresponds to the north pole, while

$$|1\rangle$$

occupies the south pole.

Advantages of the Bloch Sphere

The Bloch sphere provides

- an intuitive visualization of qubit states,
- a simple interpretation of quantum gates,
- insight into superposition,
- a convenient representation of quantum measurements,
- a useful tool for designing quantum circuits.

It is one of the most widely used graphical tools in quantum information science.

Limitations

The Bloch sphere represents only

- pure states,
- single-qubit systems.

Mixed states require the density matrix formalism, while systems containing two or more qubits require higher-dimensional Hilbert spaces and cannot be fully visualized using a single Bloch sphere.

Applications in Quantum Computing

The Bloch sphere is extensively used for

- visualization of qubit evolution,
- quantum gate design,
- quantum state tomography,
- quantum control,
- Qiskit state visualization.

Most quantum programming platforms provide built-in tools for displaying Bloch sphere representations of qubit states.

Key Takeaways

- Every pure single-qubit state corresponds to a point on the Bloch sphere.
- The angles θ and ϕ uniquely specify the qubit state.
- Quantum gates perform rotations of the Bloch vector.
- The computational basis states occupy the poles of the sphere.
- The Bloch sphere provides an intuitive picture of quantum computation.

Summary

The Bloch sphere offers a powerful geometric representation of single-qubit quantum states. By expressing a qubit in terms of two angular parameters, every pure state can be mapped to a point on the surface of a unit sphere. Quantum gates correspond to rotations of the Bloch vector, making the Bloch sphere an indispensable tool for understanding quantum state evolution, quantum circuits, and quantum algorithms. While limited to single-qubit pure states, it remains one of the most important visualization techniques in quantum computing.

Review Questions

1. What is the Bloch sphere?
2. Write the general expression for a qubit on the Bloch sphere.
3. Where are the states $|0\rangle$ and $|1\rangle$ located?
4. How are quantum gates interpreted on the Bloch sphere?
5. Why can the Bloch sphere represent only single-qubit pure states?
6. Determine the Bloch sphere coordinates of the state $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$.

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. Michael A. Nielsen, *Quantum Information Theory*.
3. David Griffiths, *Introduction to Quantum Mechanics*.
4. Jack D. Hidary, *Quantum Computing: An Applied Approach*.
5. IBM Quantum Documentation and Qiskit Textbook.

Chapter 9

No-Cloning Theorem

Learning Objectives

After completing this lecture, students will be able to

- State the No-Cloning Theorem.
- Understand why arbitrary quantum states cannot be copied.
- Explain the mathematical proof based on linearity.
- Distinguish between copying classical information and quantum information.
- Appreciate the significance of the theorem in quantum communication and cryptography.

Prerequisites

Knowledge of quantum states, Hilbert spaces, superposition, and unitary transformations.

Introduction

One of the most remarkable differences between classical and quantum information is that an unknown quantum state cannot be copied perfectly. In classical computing, files, images, and bits can be duplicated without altering the original information. Quantum mechanics, however, imposes a fundamental restriction known as the *No-Cloning Theorem*. Proposed independently by Wootters and Zurek, and by Dieks in 1982, this theorem is a cornerstone of quantum information theory and underpins the security of quantum cryptography.

Statement of the Theorem

The No-Cloning Theorem states:

> *It is impossible to construct a physical device that can create an identical copy of an arbitrary unknown quantum state.*

The theorem applies only to **unknown** quantum states. If the state is already known or belongs to a known orthogonal set, it can be prepared again using the appropriate quantum circuit.

Why Classical Information Can Be Copied

A classical bit exists in one of two definite states,

$$0 \quad \text{or} \quad 1.$$

A copying operation simply performs

$$0 \to 00, \qquad 1 \to 11.$$

Since classical information is deterministic, unlimited identical copies can be produced without violating any physical law.

Why Quantum Information Cannot Be Copied

Consider an unknown qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Suppose a universal cloning machine exists. Let the blank qubit initially be

$$|0\rangle.$$

The cloning operation would require

$$|\psi\rangle|0\rangle \longrightarrow |\psi\rangle|\psi\rangle.$$

The same machine must also satisfy

$$|0\rangle|0\rangle \to |0\rangle|0\rangle,$$

and

$$|1\rangle|0\rangle \to |1\rangle|1\rangle.$$

However, applying linearity to the superposition state gives

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle \longrightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Perfect cloning would instead require

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}.$$

These two results are different.

Therefore,

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}.$$

Hence, a universal quantum cloning machine cannot exist.

Role of Linearity

Quantum evolution is governed by unitary operators, which are linear.

If

$$U(a|\phi\rangle + b|\chi\rangle) = aU|\phi\rangle + bU|\chi\rangle,$$

then a machine capable of cloning every basis state cannot simultaneously clone every superposition state.

Thus, the impossibility of cloning follows directly from the linear structure of quantum mechanics.

Worked Example

Suppose the unknown state is

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

A hypothetical cloning machine should produce

$$|\psi\rangle|\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}.$$

Using linearity, however, the machine produces

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

which is an entangled Bell state rather than two identical copies.

Therefore, the cloning operation fails.

Applications

The No-Cloning Theorem has profound implications for quantum technologies.

- Guarantees the security of Quantum Key Distribution (QKD).
- Prevents eavesdroppers from copying transmitted qubits.
- Enables secure quantum communication.
- Motivates quantum teleportation instead of copying.
- Influences the design of quantum error-correcting codes.

Without the No-Cloning Theorem, quantum cryptographic protocols such as BB84 would not be secure.

Key Takeaways

- Unknown quantum states cannot be copied perfectly.
- The theorem follows from the linearity of quantum mechanics.
- Classical information can be duplicated, whereas quantum information cannot.
- Superposition is the key reason cloning fails.
- The theorem forms the basis of secure quantum communication.

Summary

The No-Cloning Theorem is one of the fundamental principles of quantum information science. It states that no physical process can perfectly copy an arbitrary unknown quantum state. This restriction arises naturally from the linear evolution of quantum systems and distinguishes quantum information from classical information. The theorem has far-reaching consequences in quantum cryptography, quantum communication, teleportation, and quantum computing, making it one of the cornerstones of modern quantum technology.

Review Questions

1. State the No-Cloning Theorem.
2. Why can classical information be copied but quantum information cannot?
3. Explain how linearity leads to the No-Cloning Theorem.
4. Why does cloning fail for superposition states?
5. How does the No-Cloning Theorem ensure secure quantum communication?
6. What is the difference between cloning and quantum teleportation?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
2. W. K. Wootters and W. H. Zurek, "A Single Quantum Cannot Be Cloned," *Nature*, 1982.
3. D. Dieks, "Communication by EPR Devices," *Physics Letters A*, 1982.
4. David Griffiths, *Introduction to Quantum Mechanics*.
5. IBM Quantum Documentation.

Chapter 10

Density Matrices

Learning Objectives

After completing this lecture, students will be able to

- Define the density matrix (density operator).
- Distinguish between pure and mixed quantum states.
- Construct density matrices for simple quantum systems.
- Explain the properties of density operators.
- Appreciate the role of density matrices in quantum computing.

Prerequisites

Knowledge of quantum states, Hilbert spaces, inner products, and matrix algebra.

Introduction

So far, quantum systems have been described using state vectors (kets). This description is adequate for isolated systems in pure states. However, practical quantum systems interact with their surroundings, resulting in uncertainty and loss of coherence. Such systems are better described using the *density matrix*, also called the *density operator*. The density matrix provides a unified framework for describing both pure and mixed quantum states and is widely used in quantum computing, quantum communication, and quantum information theory.

Density Operator

For a pure quantum state

$$|\psi\rangle,$$

the density operator is defined as

$$\rho = |\psi\rangle\langle\psi|.$$

The density matrix is obtained by multiplying the state vector with its conjugate transpose.

If

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

then

$$\rho = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \beta\alpha^* & |\beta|^2 \end{bmatrix}.$$

Thus, the density matrix completely describes the quantum state.

Pure States

A pure state is represented by a single normalized state vector.

For example,

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The corresponding density matrix is

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The off-diagonal elements represent quantum coherence between basis states.

Mixed States

A mixed state is a statistical ensemble of different pure states.

If a system is in state

$$|\psi_i\rangle$$

with probability

$$p_i,$$

then

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|.$$

For example, if

$$P(|0\rangle) = \frac{1}{2}, \qquad P(|1\rangle) = \frac{1}{2},$$

then

$$\rho = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Unlike the previous pure state, this density matrix contains no off-diagonal terms, indicating the absence of quantum coherence.

Properties of Density Matrices

Every valid density operator satisfies the following properties:

1. Hermitian

$$\rho^\dagger = \rho.$$

2. Unit trace

$$\mathrm{Tr}(\rho) = 1.$$

3. Positive semidefinite

   All eigenvalues are non-negative.

4. Probability interpretation

   Measurement probabilities are obtained directly from the density matrix.

These properties ensure that the density matrix represents a physically realizable quantum state.

Expectation Value

The expectation value of an observable A is calculated using

$$\langle A \rangle = \mathrm{Tr}(\rho A),$$

where Tr denotes the trace of a matrix.

This compact expression works for both pure and mixed states.

Worked Example

Consider the basis state

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Its density matrix is

$$\rho = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

The trace is

$$\mathrm{Tr}(\rho) = 1 + 0 = 1.$$

The matrix is Hermitian and has non-negative eigenvalues, confirming that it represents a valid quantum state.

Applications in Quantum Computing

Density matrices are indispensable for

- describing noisy quantum systems,
- quantum decoherence,
- quantum error correction,
- quantum communication,
- open quantum systems,
- quantum state tomography.

Modern quantum processors inevitably interact with their environment, making the density matrix formalism essential for realistic quantum computation.

Key Takeaways

- Density matrices describe both pure and mixed states.
- A pure state's density matrix is $\rho = |\psi\rangle\langle\psi|$.
- Mixed states are weighted averages of pure-state density matrices.
- Density matrices are Hermitian, positive semidefinite, and have unit trace.
- The expectation value of an observable is obtained using $\mathrm{Tr}(\rho A)$.

Summary

The density matrix extends the state-vector formalism to describe both isolated and statistical quantum systems. Pure states are represented by the outer product of a state vector with itself, while mixed states are expressed as weighted sums of pure-state density matrices. The density operator plays a central role in modern quantum computing because it naturally describes decoherence, environmental interactions, and noisy quantum devices. Consequently, it has become one of the most important mathematical tools in quantum information science.

Review Questions

1. Define the density operator.
2. How is the density matrix constructed for a pure state?
3. Distinguish between pure and mixed states.
4. State four properties of a density matrix.
5. Why are the off-diagonal elements important?
6. Write the expression for the expectation value using the density matrix.

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information.*
2. John Preskill, *Lecture Notes on Quantum Computation.*
3. J. J. Sakurai, *Modern Quantum Mechanics.*

4. David Griffiths, *Introduction to Quantum Mechanics*.

5. IBM Quantum Documentation.

Chapter 11

Mixed States

Learning Objectives

After completing this lecture, students will be able to

- Define a mixed quantum state.
- Distinguish between pure and mixed states.
- Represent mixed states using density matrices.
- Understand the physical origin of mixed states.
- Explain the importance of mixed states in practical quantum computing.

Prerequisites

Knowledge of quantum states, Hilbert spaces, density matrices, and probability theory.

Introduction

In an ideal quantum computer, every qubit exists in a well-defined pure state. In practice, however, quantum systems interact with their surroundings through noise, thermal fluctuations, and unwanted measurements. These interactions cause uncertainty in the quantum state, leading to a *mixed state*. Mixed states describe statistical ensembles of quantum states rather than a single state vector. Since real quantum processors are never perfectly isolated, mixed states provide a more realistic description of quantum systems.

Pure versus Mixed States

A **pure state** is completely described by a normalized state vector,

$$|\psi\rangle.$$

For example,

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

is a pure state.

A **mixed state** arises when the system is in one of several possible pure states with known probabilities.

For example,

$$P(|0\rangle) = 0.7, \qquad P(|1\rangle) = 0.3.$$

Here, the exact state is unknown, and only the probabilities are known.

Density Matrix Representation

Mixed states cannot be represented by a single state vector. Instead, they are described using the density operator,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|,$$

where

- p_i is the probability of the system being in state $|\psi_i\rangle$,
- $\sum_i p_i = 1$.

For the example

$$P(|0\rangle) = 0.7, \qquad P(|1\rangle) = 0.3,$$

the density matrix becomes

$$\rho = 0.7 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0.3 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.3 \end{bmatrix}.$$

The absence of off-diagonal elements indicates that the coherence between the two basis states has been lost.

Origin of Mixed States

Mixed states arise due to several physical processes:

- Interaction with the environment.
- Thermal fluctuations.
- Incomplete knowledge of the system.
- Decoherence.
- Noise during quantum operations.

Unlike classical uncertainty, quantum mixed states result from both statistical uncertainty and the loss of quantum coherence.

Purity of a Quantum State

The purity of a quantum state is measured using

$$\mathrm{Tr}(\rho^2).$$

For a pure state,

$$\mathrm{Tr}(\rho^2) = 1.$$

For a mixed state,

$$\mathrm{Tr}(\rho^2) < 1.$$

This provides a convenient criterion for distinguishing pure and mixed states.

Worked Example

Consider the mixed state

$$\rho = \begin{bmatrix} 0.6 & 0 \\ 0 & 0.4 \end{bmatrix}.$$

Its square is

$$\rho^2 = \begin{bmatrix} 0.36 & 0 \\ 0 & 0.16 \end{bmatrix}.$$

Therefore,

$$\mathrm{Tr}(\rho^2) = 0.36 + 0.16 = 0.52.$$

Since

$$0.52 < 1,$$

the state is a mixed state.

By comparison, any pure state's density matrix satisfies

$$\mathrm{Tr}(\rho^2) = 1.$$

Mixed States and Decoherence

Quantum coherence enables superposition and interference. When a quantum system interacts with its surroundings, coherence gradually disappears, a phenomenon known as *decoherence*.

Decoherence transforms pure states into mixed states, limiting the performance of quantum computers. Considerable research is devoted to reducing decoherence through improved hardware, cryogenic cooling, and quantum error correction.

Applications in Quantum Computing

Mixed states play a vital role in

- modeling noisy quantum processors,
- quantum error correction,
- quantum communication,
- open quantum systems,
- quantum cryptography,
- quantum state tomography.

Accurate simulation of practical quantum devices requires the density matrix formalism rather than ideal state vectors.

Key Takeaways

- Mixed states represent statistical ensembles of quantum states.
- They are described using density matrices.
- Environmental interactions convert pure states into mixed states.
- The purity condition $\mathrm{Tr}(\rho^2)$ distinguishes pure and mixed states.
- Mixed states provide a realistic description of present-day quantum hardware.

Summary

Mixed states arise whenever a quantum system cannot be described by a single state vector due to uncertainty or interaction with its environment. They are represented by density matrices, which combine multiple pure states according to their probabilities. The quantity $\mathrm{Tr}(\rho^2)$ serves as a measure of purity, with pure states satisfying $\mathrm{Tr}(\rho^2) = 1$ and mixed states satisfying $\mathrm{Tr}(\rho^2) < 1$. Because real quantum computers are affected by noise and decoherence, mixed-state analysis is indispensable in modern quantum information science.

Review Questions

1. Define a mixed quantum state.
2. How does a mixed state differ from a pure state?
3. Write the density matrix for a mixed state.
4. What physical processes produce mixed states?
5. What is the significance of $\mathrm{Tr}(\rho^2)$?
6. Why are mixed states important in practical quantum computing?

References

1. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information.*
2. John Preskill, *Lecture Notes on Quantum Computation.*
3. J. J. Sakurai, *Modern Quantum Mechanics.*

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